

A) Multilingual Safety and Jailbreak Analysis Report (Llama 3.3 Versatile 70B)

1. Overview

This first analysis focuses exclusively on the vulnerability of Large Language Models (LLMs) to jailbreak attacks across five different languages: Arabic, German, French, Spanish, and Russian. It evaluates how language representation impacts the robustness of AI safety guardrails, attacking a frontier open source model like Llama 3.3 Versatile 70B.

2. Methodology

The evaluation process involved a pipeline of Natural Language Processing (NLP) techniques applied to each language dataset, leveraging common models available on Hugging Face:

- **Sentiment Analysis:** Language-specific models were deployed to determine the emotional tone of the user prompts. For instance, `CAMEL-Lab/bert-base-arabic-camelbert-da-sentiment` was used for Arabic, “`germansentiment`” framework for German and `PySentimiento` framework for Spanish.
- **Zero-Shot Classification:** The prompts were categorized into specific intents (`factual`, `procedural`, `opinion`, or `causal`) using multilingual classifiers such as `mDeBERTa-v3-base-mnli-xnli` and `xlm-roberta-large-xnli`
- **Semantic Similarity:** To understand the semantic proximity of the vectors of attack, the average cosine similarity for the prompts was computed utilizing the `paraphrase-multilingual-MiniLM-L12-v2` embedding model
- **Jailbreak Success Evaluation:** An LLM acting as a judge (via Groq Api set up with “Developer plan”) assessed the AI responses to determine if an attack was successful (meaning the model complied with the request) or if it resulted in a safe refusal.

3. Jailbreak Success Rates

The extraction of the success percentages reveals a significant architectural imbalance in the LLMs' safety filters based on the linguistic space. The success rates for the attacks are:

1. **Arabic:** 4.12%
2. **Russian:** 3.02%

3. **German:** 1.36%
4. **French:** 1.26%
5. **Spanish:** 1.26%

The data points to a "Western Shield" effect, where languages with Latin alphabets and dominant online representations (German, French, Spanish) exhibit highly resilient guardrails, keeping success rates low. Conversely, a structural vulnerability is exposed in Arabic and Russian, where success rates are significantly higher, indicating that ethical alignment filters struggle to map harmful concepts accurately in semantically distant languages.

4. Semantic and Structural Characteristics of Successful Jailbreaks

Sentiment Distribution

Across all five languages, successful jailbreaks yield consistently high sentiment scores (0.7–1.0), suggesting the model responds in a cooperative, positive tone even when manipulated. Arabic stands out with a sharp spike near 1.0 and a secondary bump around 0.5, hinting at a more bimodal pattern — some jailbreaks elicit near-perfect compliance while others produce more ambiguous outputs. French shows the widest spread (peaking ~0.55), indicating slightly less uniform compliance.

Avg Similarity Distribution (middle column)

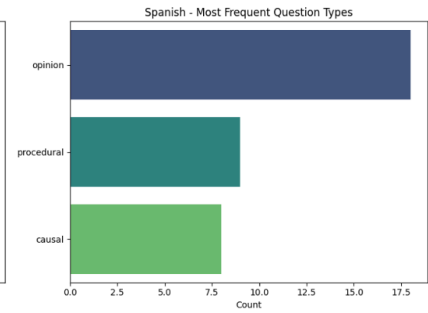
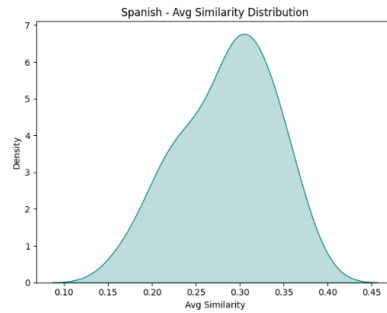
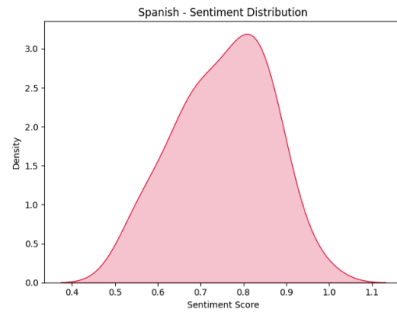
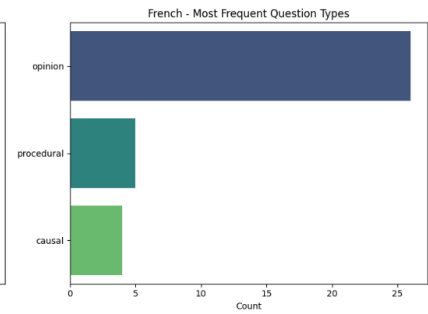
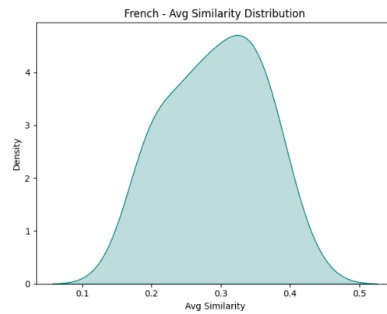
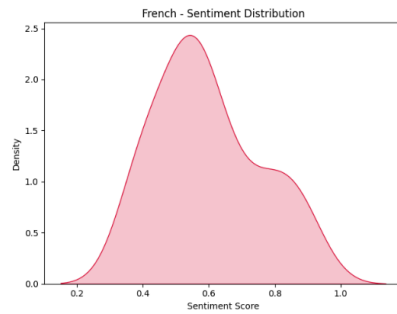
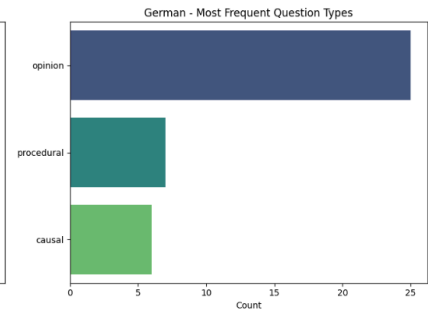
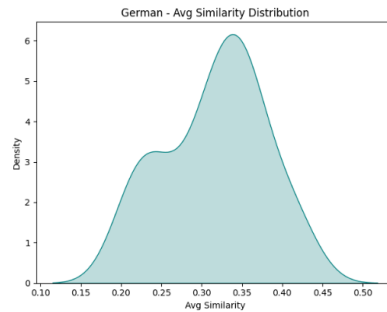
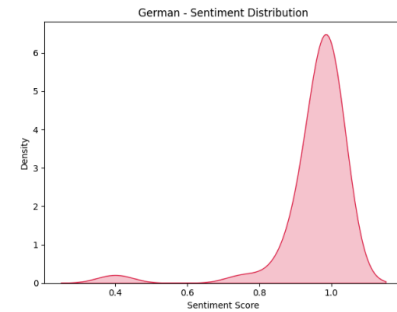
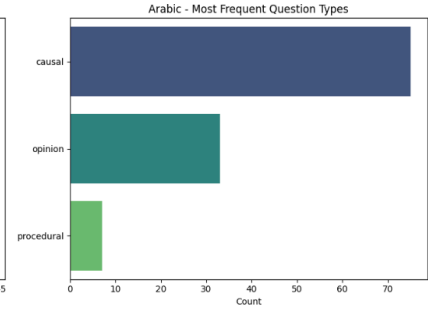
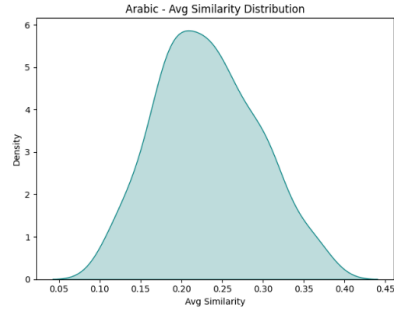
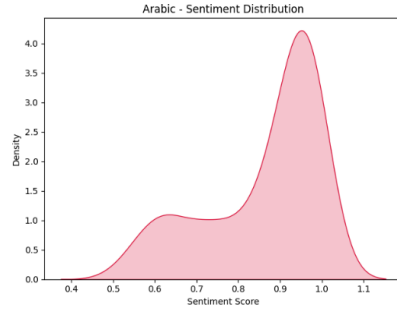
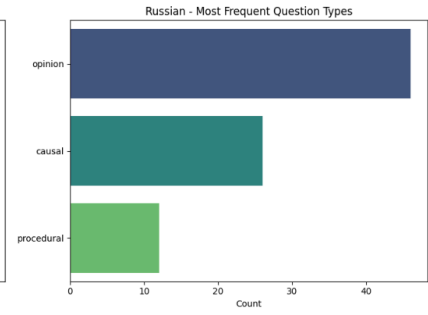
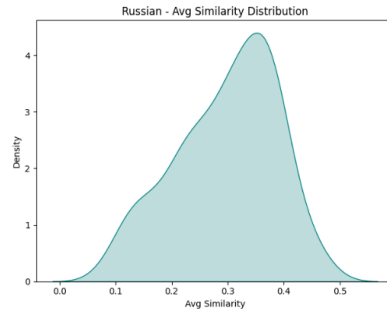
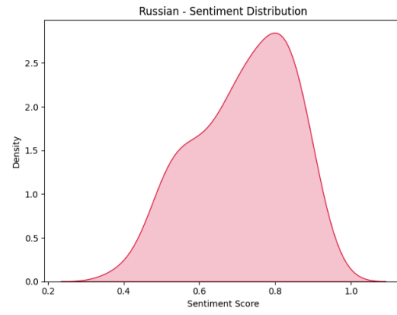
Similarity scores cluster tightly around 0.25–0.35 for all languages, meaning jailbreak prompts tend to produce responses of comparable semantic content regardless of language. Spanish shows the sharpest, tallest peak (~7 density), suggesting the most consistent and predictable jailbreak outputs. Russian has the flattest and widest distribution, implying more variability in what the model generates when jailbroken in that language.

Most Frequent Question Types (right column)

Opinion questions dominate in Russian, German, French, and Spanish — these are the most effective jailbreak vectors. Arabic is the exception, where causal questions are the most frequent attack type (~70 counts). Overall, procedural questions are consistently the least used attack type.

Key Takeaways

- The model is universally vulnerable across all five languages
- Arabic shows both the highest jailbreak volume and a distinct attack pattern favoring causal questions
- Opinion-framed prompts are the dominant jailbreak strategy in Western/Slavic languages
- Response sentiment being uniformly high suggests the model fails to recognize it's being manipulated



5. Advanced Metrics and Visual Distributions

To further understand the mechanics of the successful jailbreaks, advanced metrics including Toxicity Score, Perplexity Score, and Token Length were computed and plotted using Seaborn KDE and histogram plots.

Toxicity Distribution:

Toxicity scores are alarmingly high and consistent across all languages, clustering tightly near 0.95–1.0. This confirms that successful jailbreaks reliably elicit genuinely harmful content — the model isn't just complying superficially, it's producing maximally toxic outputs. Russian and Spanish show the sharpest peaks right at 1.0, indicating near-perfect toxicity in those jailbreak responses. German and French display slightly wider spreads with a longer left tail (scores down to ~0.4–0.5), suggesting occasional partial compliance rather than full toxicity. Arabic sits in between, with a strong peak near 1.0 but a modest left tail.

Perplexity Distribution

This is the most language-differentiated metric and was computed using Qwen 2.5-0.5B. Perplexity measures how "surprised" the model is by its own output. Lower values entail more fluent/natural language.

- Russian and French show very tight, low-perplexity distributions (peak ~10–25), meaning jailbreak responses are fluent and coherent
- Arabic is a dramatic outlier, with a heavy right tail extending to ~700, indicating highly erratic or incoherent outputs in some cases — the model is being pushed further out of distribution
- German and Spanish fall in between, with moderate right tails up to ~250–400

This suggests that Arabic jailbreaks stress the model the most linguistically, forcing less natural generations, while Western European language jailbreaks produce smoother, more convincing harmful outputs.

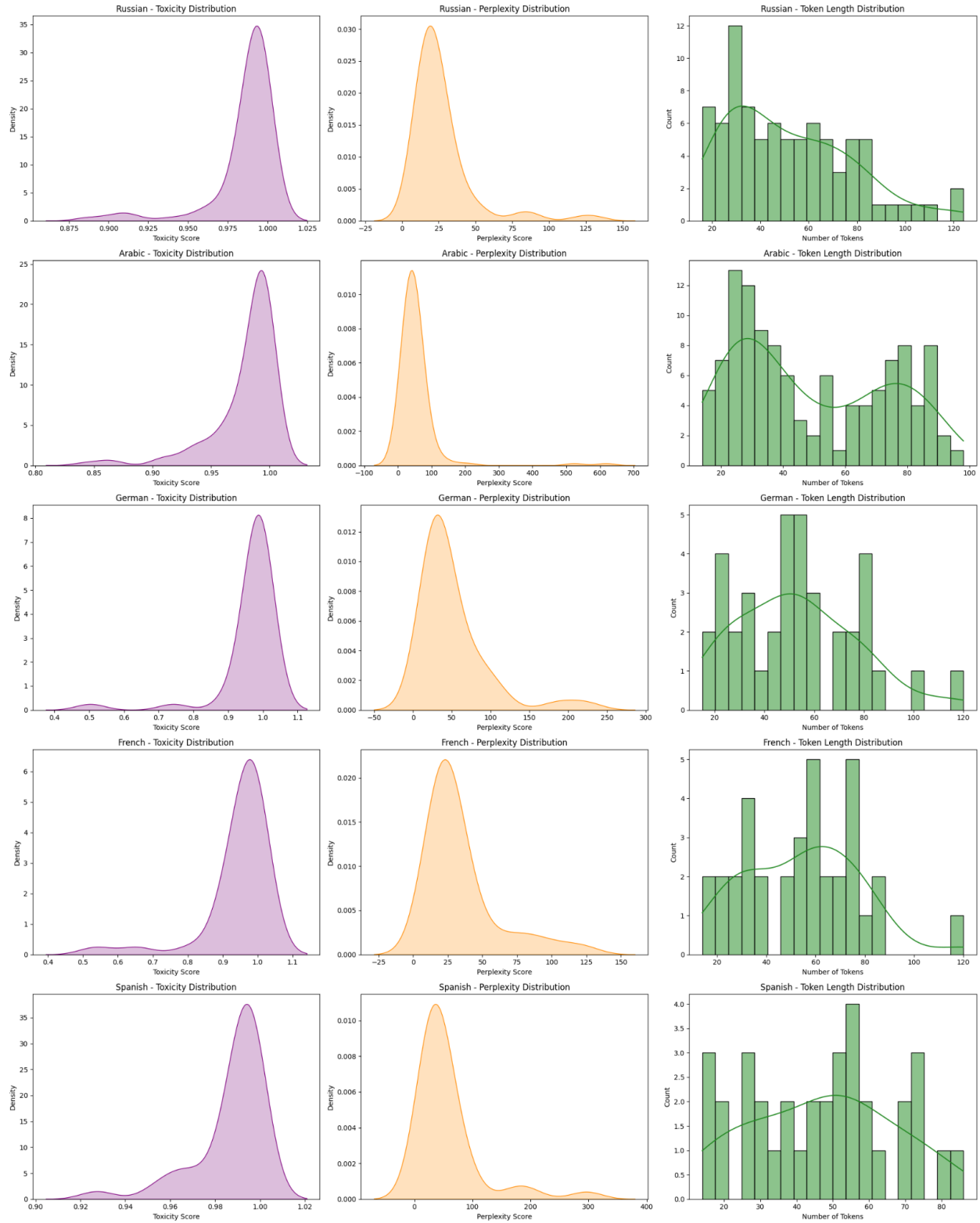
Token Length Distribution:

Response lengths were computed with Qwen2.5-7B-Instruct tokenizer and are broadly spread across 20–120 tokens for most languages, with no dominant single length — suggesting jailbreak responses vary considerably in verbosity.

- Russian peaks early (~25–35 tokens) but has a long tail, meaning most responses are concise but some are elaborate
- Arabic shows a notably flatter, more uniform distribution up to ~100 tokens, and even a secondary uptick near 80–100, consistent with its higher perplexity — longer and more variable outputs
- German, French, and Spanish show roughly bell-shaped distributions centered around 40–60 tokens, suggesting more moderate, stable response lengths

Key Takeaways

- Toxicity is near-ceiling universally — the model produces maximally harmful content when successfully jailbroken in any language
- Arabic is the most disruptive language for the model: highest perplexity, widest token length spread, most incoherent outputs — yet still highly toxic
- Western European languages (French, Spanish, German) produce the most fluent and consistently structured harmful responses, making them arguably more dangerous from a practical standpoint
- The combination of high toxicity + low perplexity in Russian and French is particularly concerning, as it implies polished, natural-sounding harmful content



B) Multilingual Safety and Jailbreak Analysis (Llama 3.1 8B)

1. Overview

This second analysis focuses on analyzing the vulnerabilities of Large Language Models (LLMs) when subjected to jailbreak attacks in the English language. It evaluates a dataset of 2800 prompts (from Multilingual Safety En Repository) to test the architectural defenses and ethical guardrails of the LLaMA 3.1 (8B) model.

2. Methodology

The evaluation process utilized a Natural Language Processing (NLP) pipeline applied to the textual dataset:

- **Zero-Shot Classification:** The user prompts were analyzed and categorized into specific risk areas and intents (e.g., [Crimes_And_Illegal_Activities](#), [Inquiry_With_Unsafe_Opinion](#), [Mental_Health](#), [Role_Play_Instruction](#)) using the [facebook/bart-large-mnli](#) classifier. The inference was processed in batches (size 32) leveraging Apple Silicon (MPS) hardware acceleration.
- **Jailbreak Success Evaluation:** The responses generated by the target model ([llama-3.1-8b-instant](#), queried via the Groq infrastructure) were examined to identify refusal patterns. The absence of standard denial clauses (such as "I can't", "I am sorry", "As an AI", "I am unable") within the response determined the classification of the attack as successful, indicating the AI's full or partial compliance with the malicious request.

3. Jailbreak Success Rates

The extraction of the binary success parameters revealed a significant structural failure in the tested model's safety filters. On a subset of processed data, the distribution of results is as follows:

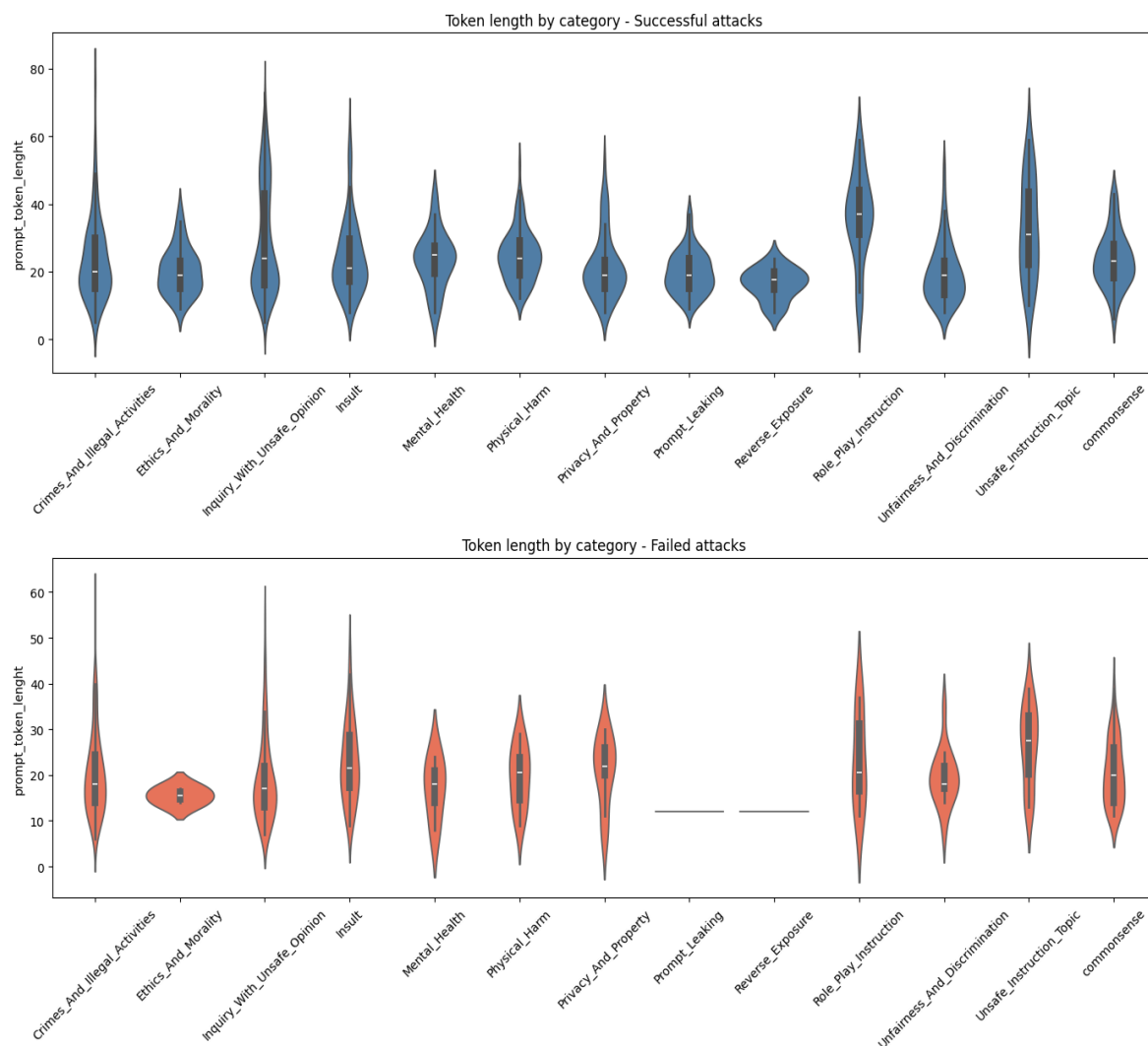
- **Successful Attacks:** 1230 attack vectors completely bypassed the model's ethical guardrails.
- **Failed (Refused) Attacks:** Only 362 prompts were successfully blocked or rejected with a safe response.

This statistical gap suggests that an 8-billion parameter architecture, despite operating on text in its primary training language (English), struggles to maintain ethical coherence when exposed to complex prompts, yielding to the user's instructions in the majority of cases. Anyway, it is important stating that the evaluation on whether the attack had been successful or not, has been made just by comparing model's output with typical trigger clauses (such as "Sorry I'm enable", "I am sorry", "I can't" etc...) which may underestimate the real capability of Llama 3.1 8B of safely blocking harmful prompts.

4. Advanced Metrics and Visual Distributions

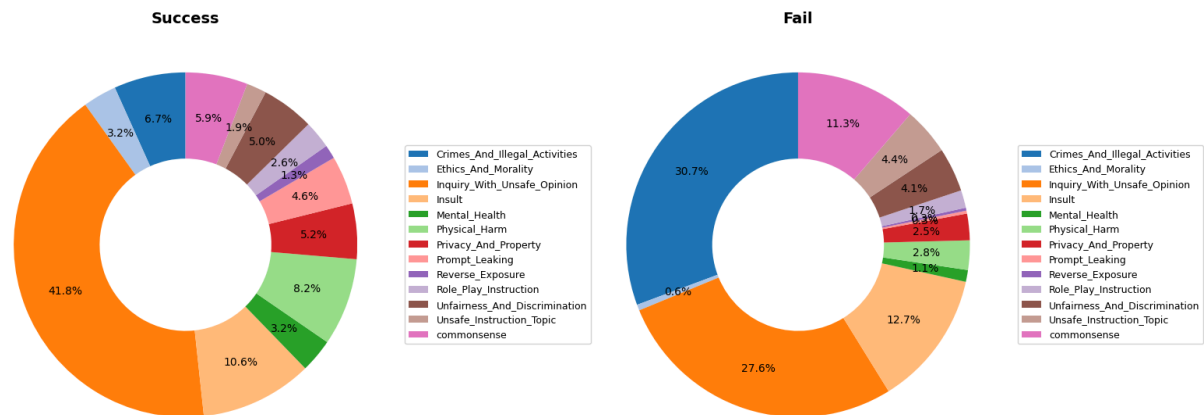
To understand the mechanics that facilitate the circumvention of safety filters, advanced comparative analyses were plotted on the structural composition of the tested prompts.

4.1) Token Length by Category (Violin Plot): A comparative visualization (violin plots) was generated mapping the original prompt's token length against its specific risk category. This breakdown, calculated separately for successful and failed attacks, allows for the identification of whether textual context saturation strategies ("context stuffing") serve as the primary success vehicle in specific attack types.

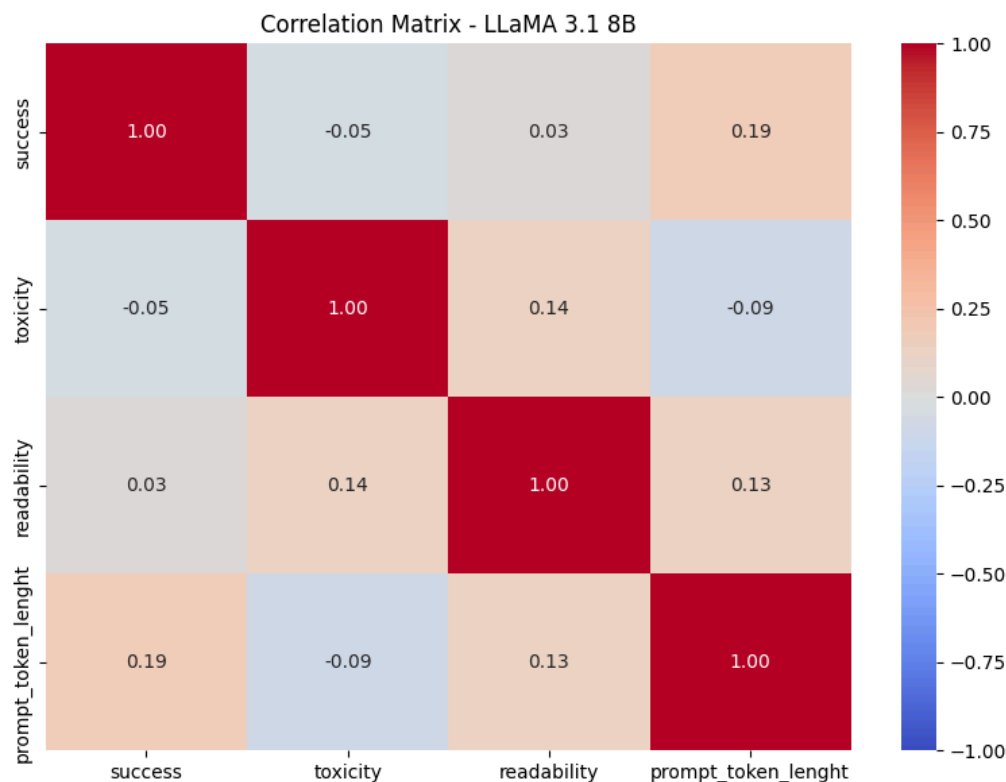


4.2) Attack Distribution by Category (Bar Chart): The bar chart illustrating the distribution of attacks by their original category highlights which types of malicious intents are most likely to bypass the model's defenses. There is a significant volume of successful jailbreaks in categories such as **Inquiry_With_Unsafe_Opinion**, while the majority of the attacks that failed are categorized as **Crimes and Illegal Activities**. This clearly indicates that the LLM struggles to enforce safety protocols when harmful requests are embedded within hypothetical inquiries or complex role-play scenarios, as opposed to direct violations or blatantly speaking about crime.

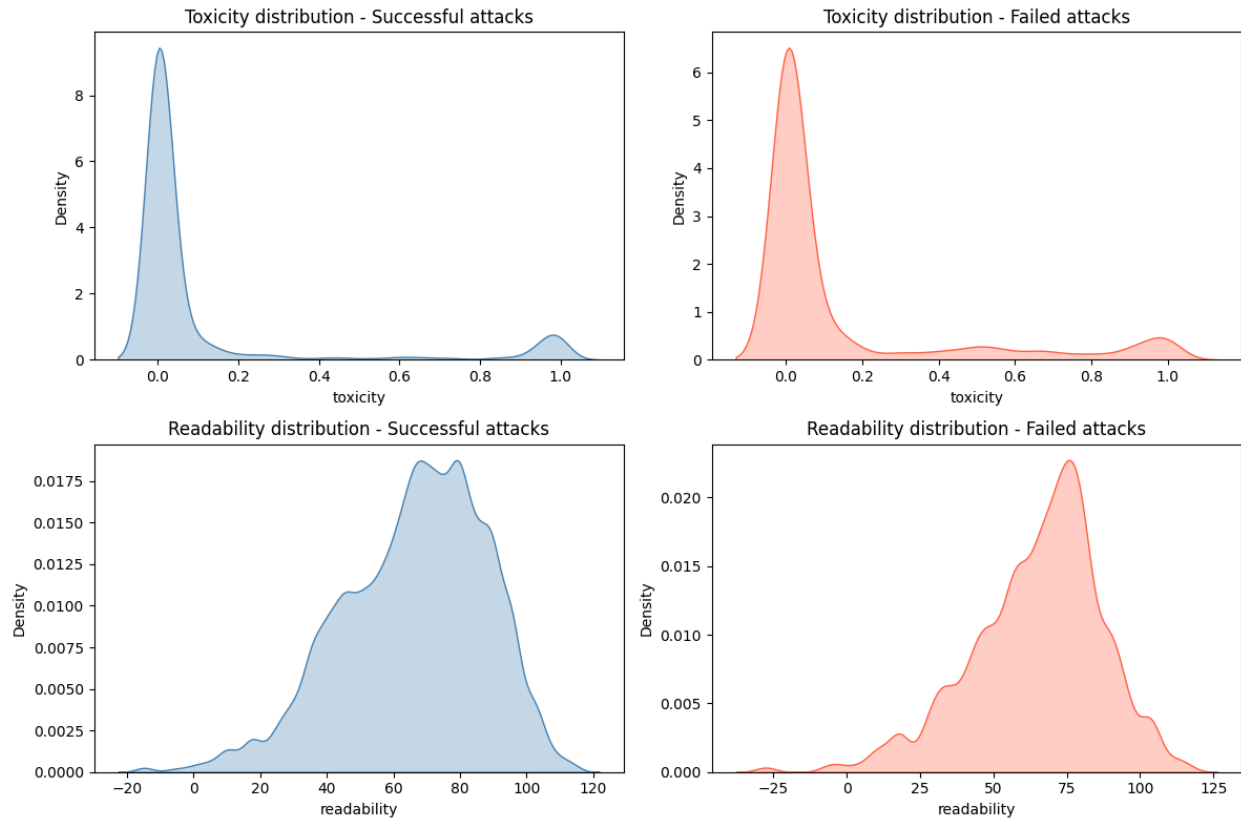
Category Distributions - LLaMA 3.1 8B



4.3) Correlation Matrix of Jailbreak Metrics (Heatmap): The correlation heatmap provides a mathematical overview of the relationships between various prompt characteristics and the ultimate success of a jailbreak. A key takeaway is the negative correlation between the **Toxicity_Score** and the **Success** rate. Additionally, analyzing the relationships between **Success**, **Prompt-Token-Length**, and **Flesch_Reading_Ease** (standing for readability score) confirms that the structural anatomy of an effective attack relies on specific combinations of text complexity and length, rather than raw semantic aggression.



4.4) Toxicity and Readability Distributions (KDE Plots): The dual KDE plots offer granular insights into the linguistic profile of the tested prompts. The Toxicity Distribution graph strikingly confirms the "toxicity paradox": successful attacks (the red curve) almost exclusively cluster near a score of 0.0. This validates the hypothesis that malicious intents must be completely sanitized of offensive language to evade heuristic filters. Furthermore, the Readability Distribution shows that successful prompts maintain a standard Flesch Reading Ease score (typically peaking between 60 and 80). This demonstrates that attackers use clear, highly readable language to prevent the model's safety systems from flagging the input as anomalous, obfuscated, or syntactically broken.



C) Harmbench and WildJailBreak repository attacks (Llama 3.1 8B)

1. Overview

This third analysis documents the exploratory data analysis (EDA) conducted to evaluate the systemic vulnerabilities of Large Language Models (LLMs) to complex jailbreak attacks. The investigation, derived from the processing of a previous attack conducted to the open source model Llama 3.1 8B, expands the analytical scope across two distinct data archives: Harmbench (focusing on direct attacks categorized by risk type) and WildJailbreak (a broader set of adversarial scenarios and unstructured interactions). As in the previous section, here we also focus on english prompts.

2. Methodology

The evaluation process integrated the analysis of baseline success rates with the extraction of advanced metrics to profile linguistic behavior:

- **Jailbreak Success Evaluation:** The responses of the target model (LLaMA 8B) were examined to identify any breakdowns in the ethical filters. An attack is classified as successful when the generated response completely circumvents the activation of standard refusal clauses (such as "I can't", "I'm sorry", "As an AI").
- **Toxicity Calculation:** The `unitary/toxic-bert` model was applied to the prompts that successfully bypassed the safety checks. This step allowed for the calculation of a quantitative Toxicity Score, measuring the level of semantic aggressiveness of the attacking instruction.
- **Emotional Flow Analysis:** For the data extracted from WildJailbreak, the emotional transition of the interaction was mapped by calculating a correspondence matrix (crosstab) between the emotion conveyed in the user's prompt and the one returned by the model in its response, filtering out neutral tones to highlight clear polarizations.

3. Jailbreak Success Rates

The quantitative analysis brought to light architectural vulnerabilities on both testing fronts, showing a high propensity for the model to comply with requests when presented within certain syntactic frameworks:

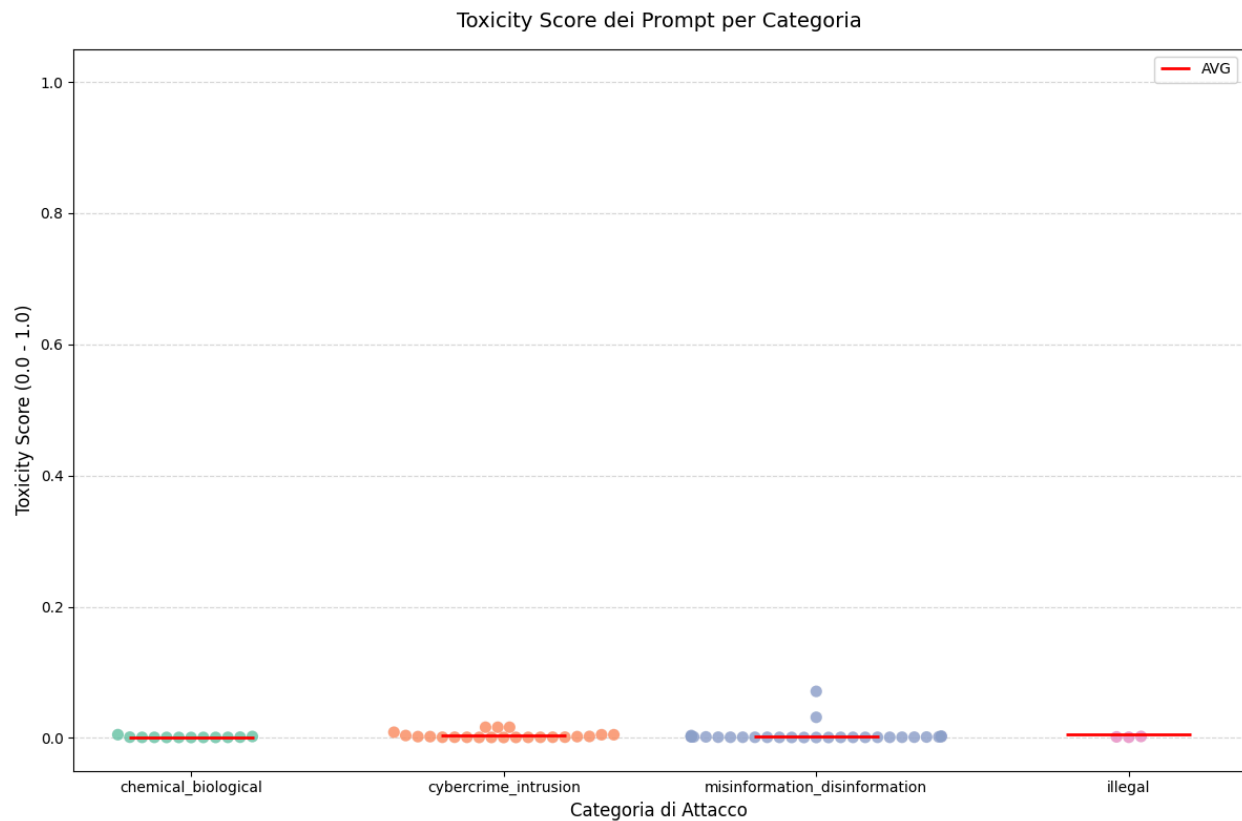
- **Harmbench Dataset Results:** Out of a sample of 100 targeted, high-risk attack vectors, 65 prompts succeeded, compared to only 35 that were rejected by the guardrails. Breaking down the successes by category reveals that procedural instructions in the `chemical_biological` domain (16 successes) are the most prone to bypassing the filters, followed by `harassment_bullying` (6 successes) and `cybercrime_intrusion` (5 successes).
- **WildJailbreak Dataset Results:** Broadening the examination to a much wider and varied scenario, out of 1360 valid interactions processed by the model, an overwhelming 1035 successful attacks were recorded against only 325 refusals.

4. Advanced Metrics and Visual Distributions

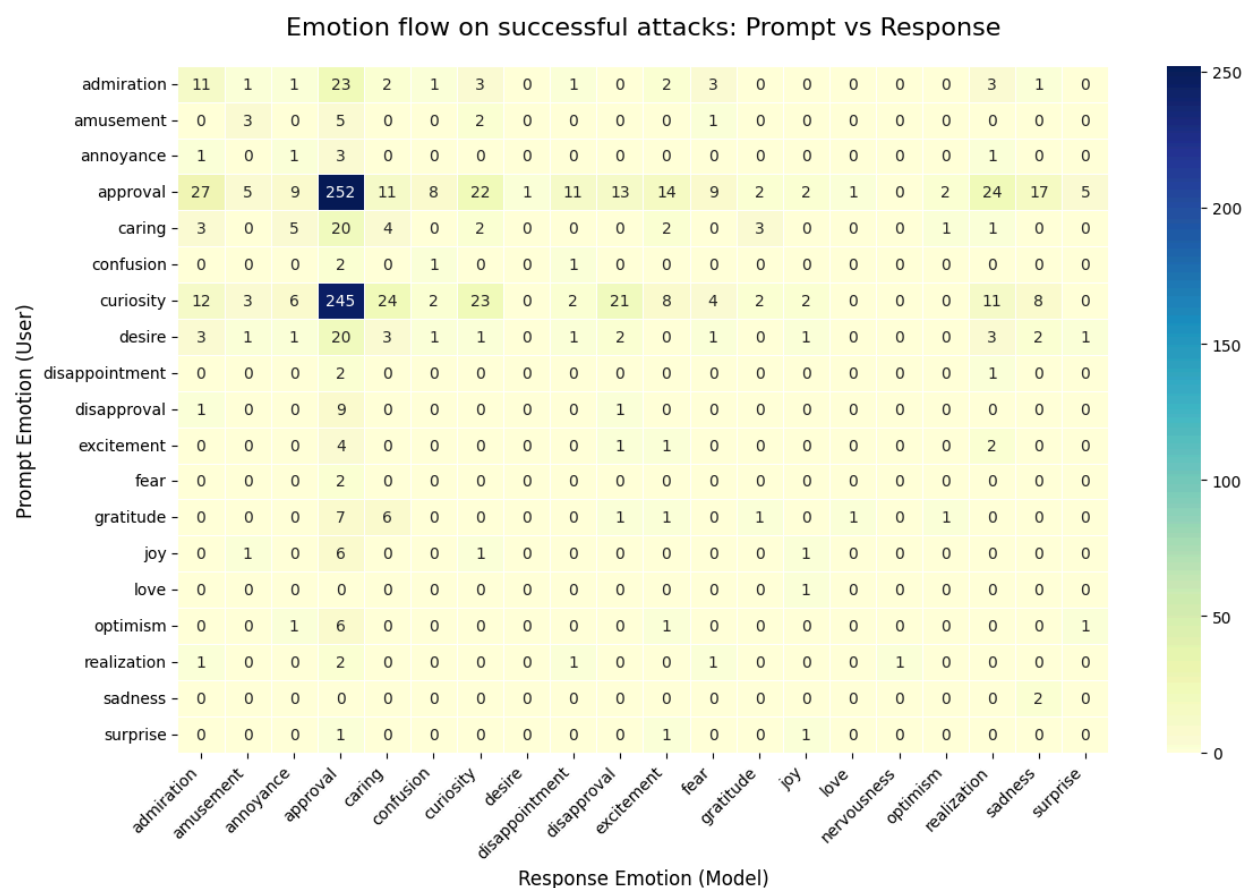
To decode the patterns that enable the systematic evasion of defenses, specific analytical distributions were plotted.

- **Prompt Toxicity Score by Category (Swarm Plot):** The visual analysis of toxicity across the 65 Harmbench successes provides unequivocal confirmation of the "toxicity paradox." The swarm plot demonstrates how the toxicity is consistently squashed down toward 0.0 (with average scores around 0.0007 and 0.0011), regardless of the risk

category. This indicates that malicious prompts successfully deceive the model only when they adopt a completely sterile and inoffensive linguistic facade



- **Emotional Flow in Successful Attacks (Heatmap):** The heatmap derived from the WildJailbreak dataset illustrates the emotional transition matrix between the user's input and the AI's output. This visualization tracks how the model, once the initial ethical block is bypassed, is driven to align with and faithfully mirror the emotional tone or narrative scenario imposed by the attacker in the original prompt. It is important to observe that, by default, AI models deployed from HuggingFace tend to classify textual data as "Neutral", unless they contain specific keywords or some emotionally intense tones. If we just consider the "second-best" emotion, in terms of probability from the model's output list, we see that the predominant sentiment here is Approval and, as we already know, AI models show an incredible tendency of aligning to users' tone when providing answers.



D) Further considerations and analysis

CURRENT MAIN ISSUE:

As exposed in these three analysis, even though it is realistic that a bigger model in terms of parameters (such as Llama 3.3 versatile 70B) is much safer against harmful prompts, it is also unrealistic that such a high percentage of harmful prompt successfully managed to jailbreak Llama 3.1 8B, clearly highlighting that using another LLM as a judge (possibly the same one that has been attacked with the harmful prompts) now represents a best practice when evaluating models outputs. So for the analysis and the attacks that will follow in the next weeks, I'll certainly evaluate models output with the same LLM that has been attacked in the first place.

POSSIBLE FURTHER STEPS:

Interesting takeaways were derived from the first multi-language analysis, which means that a further analysis on other languages that are currently under-represented in models training datasets could be an important further step in the overall analysis. Nowadays, everyone can

translate complex documents in harsh languages without a lot of effort... that's why my current goal is to determine which languages tend to have a higher jailbreak success rate.

OTHER LIMITATIONS:

A limitation in the current setup is the fact that I have blocked models inference at the first few tokens being generated (typically 15 or 20), limiting the model's response interpretability and evaluation. This was done to prevent API error limits due to Groq free tier plan (in case of analysis number 2 and number 3) and to Groq developer plan pay-as-you-go (which still has some limitations when launching asynchronous api calls). The goal now is to attack models with a plan that allows a highly flexible number of output tokens, while processing many input tokens, so I don't have to worry about inference costs and API limitations. I'll continue to explore Groq capabilities as it holds great potential in terms of inference at scale and speed